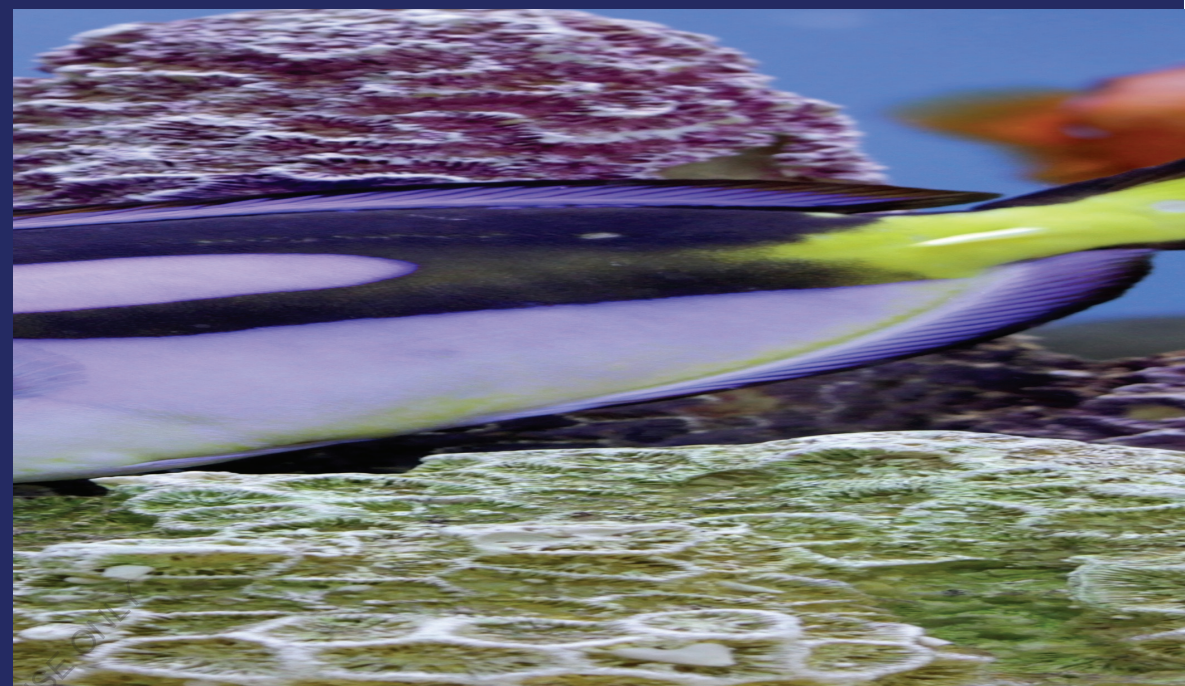


This book adheres to the scope and sequence of most major aquarium-keeping courses nationwide. It also strives to make aquarium keeping an interesting and accessible discipline for students. In addition to providing comprehensive coverage of core concepts and foundational research, it incorporates features that engage learners in meaningful ways.

This book helps teachers develop a shared vision and understanding of the basic concepts and techniques of aquarium keeping, as well as their flexible application across subjects and grade levels. It has been created with several goals in mind: accessibility, customization, and student engagement, while encouraging high levels of academic scholarship. Both instructors and students will find it to be a strong introduction to aquarium keeping in an accessible and practical format.



Dr. S. Kalaimani is a biological researcher specializing in various fields of zoological studies and serves as an Assistant Professor of Zoology at J.K.K. Nataraja College of Arts and Science, Komarapalayam, Namakkal District. He has 14 years of experience as a postgraduate Biology teacher in schools and colleges, along with research experience.



Kalaimani Subramani
Mohan K.
Revathi N.

The Book of Aquarium Keeping

Preparatory book can be used for UG and PG
Zoology students



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THE BOOK OF AQUARIUM KEEPING

First Edition-2026

Authors:

Dr. S. KALAIMANI., M.Sc., M.Phil., Ph.D., M.Ed., SET(L.sci), SET(E.du),

Assistant Professor

PG and Research Department of Zoology

J.K.K. Nataraja College of Arts & Science (Autonomous)

Komarapalayam, Namakkal D.t, Tamil Nadu, India.

Dr. K. MOHAN., M.Sc., M.Phil., Ph.D.,

Assistant Professor

PG and Research Department of Zoology,

Sri Vasavi College, Erode, Palani - Tamilnadu, India.

Cell: 9080656422

Dr. N. REVATHI., M.Sc., M.Phil., Ph.D.

Assistant Professor

PG & Research Department of Zoology

Sri Vasavi College

Erode-638316

Dr. M. ELANCHEZHIAN., M.Sc., M.Ed., M.Phil., Ph.D.,

Assistant Professor

Department of Zoology,

Erode Arts and Science (Autonomous) Erode-638012, Tamil Nadu, India.

FOREWORD

The book adheres to the scope and sequence of most one major course nationwide. also strive to make Aquarium keeping as a discipline, interesting and accessible to students. in addition to a comprehensive coverage of core concepts and foundational research, incorporated features that draw learners into the discipline in meaningful ways.

This book provides a fundamental concept of Aquarium keeping with a scope that is manageable for instructors and students alike with this; it is my pleasure to write a brief foreword to this wonderful preparatory book on "The Book of Aquarium keeping".

I am appreciating the authors to chosen this book chapter on basic preparatory knowledge of Principles of Aquarium keeping. on the aspect of Aquarium keeping include number of books are available in the form of hand book, question bank for competitive examinations etc. except from this for the student community limited numbers of this like book are available in UG or PG level lack of syllabus covered text books are available, this book might be useful to basic and advanced learner of Aquarium keeping. written of this contain simple language and easy to understand a basic learner and valuable guide to the intellectual.

This book can help teachers develop a shared vision and understanding of basic concepts of Aquarium keeping and techniques of biology and its flexible uses across subjects and grade levels.

This book has been created with several goals in mind, accessibility, customization, and student engagement all while encouraging students toward high levels of academic scholarship. instructors and students alike will find that a strong introduction to Aquarium keeping in an accessible format.



Dr. S. UMAMAHESWARI
Assistant Professor of Zoology
J. K. K. Nataraja College of Arts and Science (Autonomous)
Komarapalayam-638183.

PREFACE

A Students who has just entered the portals of colleges finds it difficult to understand the subject taught to him/her this difficulty is mainly due to understand standard references. While preparing this “**THE BOOK OF AQUARIUM KEEPING**”. has been prepared to serve the special purpose of college students.

This book explains the aquarium keeping in a simplified manner. This Preparatory book can be used for UG and PG Zoology students, who are looking for a quick revision before the Examination.

This book support to the Zoology students to get confidence and also reference just before attending University Examination and other Competitive Examination.

This book has been written to Students and preparatory the field of Life sciences, no such book which cover suitable original picture with needed explanation Aquarium Keeping techniques and Practice.

This book is useful for the preparation of state Universities and College students integrated Certificate and Diploma courses field of Aquarium keeping.

**Dr. S. KALAIMANI.,
M.Sc.,M.Phil.,Ph.D.,M.Ed.,SET(L.sci), SET(E.du),**

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Words find no bounds to express my sincere thanks to all who made this Book to come true. my sincere thanks to **Dr.N. NALINI**, Captain M.Sc., MPhil., Ph.D. Principal, JKKN college Arts and science (Autonomous), Komarapalayam, Namakkal D.t, for extending the facilities to undergo this Book work.

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My express my deep sense of feeling of my affectionate father **P, SUBRAMANI**, my mother **Mrs.S,VEERAMMAL**, and my sister **Mrs. RUBY** and my brother **Mr. VENICE**, to whom ,I owe everything for their dedicated efforts and loving encouragement to complete this book successfully.

Last but not least i wish to extend my sincere thanks to all my friends and relatives for their co-operation and support in this book to make success.

**Dr. S. KALAIMANI.,
M.Sc.,M.Phil.,Ph.D.,M.Ed.,SET(L.sci), SET(E.du),**

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UNIT I

AQUARIUM FISH KEEPING AS HOBBY AND BUSINESS

Aquarium fish keeping is the practice of maintaining aquatic environments freshwater, brackish, or marine for the care, display, breeding, and sometimes sale of fish, invertebrates, and plants. As a hobby, it blends science (water chemistry, biology), design (aqua scaping, lighting), and routine husbandry. As a business, it applies those same principles at scale with added layers of regulation, finance, logistics, and customer service.

SCOPE OF AQUARIUM FISH KEEPING

1. ENVIRONMENTS COVERED

- Freshwater: community tanks, planted aquaria, species-specific setups (cichlids, bettas).
- Brackish: estuary-simulating systems and tolerant species.
- Marine: fish-only, mixed reef (fish + corals), SPS/LPS coral-focused systems.
- Aquaculture/RAS: intensive recirculating systems for production and commercial breeding.

2.CORE TECHNICAL DOMAINS

- Water chemistry and cycles (pH, ammonia–nitrite–nitrate, GH/KH, salinity, dissolved oxygen, temperature).
- Filtration and flow (mechanical, chemical, biological; sumps; protein skimmers).
- Lighting and photosynthetic requirements (plants, corals).
- Nutrition and feeding strategies.
- Disease prevention, quarantine, diagnosis, and treatment.
- Breeding and propagation (fish and corals).
- Aquascaping, substrate, and habitat design.
- Monitoring and automation (sensors, controllers, dosing, ATO).

3.PRACTICAL HOBBY ACTIVITIES

- Planning and cycling a tank; selecting compatible species.
- Regular maintenance: water tests, partial changes, filter care, algae control.
- Aquascaping and aesthetic care.
- Breeding as a learning activity and species conservation.
- Community engagement (clubs, forums, content creation).

4.BUSINESS ACTIVITIES AND MODELS

- Retail (brick-and-mortar stores selling livestock, supplies).
- Online sales (live fish, corals, plants, equipment).
- Breeding farms and wholesale supply.
- Maintenance services (regular tank care for homes, offices, restaurants).
- Aquascaping/design and installation contracts.
- Education, workshops, and content monetization.

5.COMPLIANCE, ETHICS, AND BIOSECURITY

- Legal requirements: local business licensing, animal welfare laws, import/export and CITES restrictions.
- Biosecurity: quarantine protocols, disease reporting, safe disposal of biological waste.
- Ethical sourcing and avoiding invasive/illegal species trade.

6.BUSINESS MANAGEMENT ESSENTIALS

- Market research, business model selection, startup and operating costs.
- Inventory and health warranties, supplier relations.
- Pricing, cash flow, P&L forecasting, break-even analysis.
- Marketing, customer service, and aftercare support.

7.SKILLS AND RESOURCES NEEDED

- Technical: water testing, filtration management, disease ID, breeding techniques.
- Operational: inventory control, basic plumbing/electrical, small-animal transport.
- Business: bookkeeping, marketing, regulatory compliance, customer relations.
- Tools: test kits, refractometer, PAR meter, quarantine tanks, RAS components, CRM/accounting software.

8.OUTCOMES AND GOALS

- Hobby: personal enrichment, aesthetic satisfaction, species conservation, learning biology/ecology.
- Business: sustainable revenue from ethical livestock and services, scalable operations, strong customer trust and compliance.

COMMERCIAL ASPECTS: NATIONAL AND INTERNATIONAL MARKET

- Commercial aspects refer to the business and economic factors involved in producing, marketing, distributing, and selling goods and services.
- They determine the profitability and growth of an industry or business.
- Commercial aspects can be broadly classified into **national (domestic) market aspects** and **international (global) market aspects**.

1. NATIONAL MARKET (DOMESTIC MARKET)

A national market refers to the buying and selling of products and services within the boundaries of a country. It focuses on meeting the demand of domestic consumers and industries.

a) Market Demand and Supply

- Demand depends on consumer preferences, income levels, population growth, and purchasing power.
- Supply depends on production capacity, availability of raw materials, labor, and technology.
- The balance between demand and supply influences market stability and growth.

b) Consumer Preferences and Market Trends

- Consumer tastes and preferences vary according to age, income, culture, and lifestyle.
- Businesses conduct market research to understand changing trends and customer needs.
- Product quality, packaging, and branding influence consumer behavior.

c) Pricing and Competition

- Prices are affected by production costs, taxes, demand, and competition.
- Competition among local producers encourages innovation and quality improvement.
- Businesses adopt pricing strategies to attract consumers and increase market share.

d) Distribution and Marketing Channels

- Products reach consumers through wholesalers, retailers, online platforms, and direct sales.
- Efficient transportation and communication systems improve market accessibility.
- Advertising and promotional activities create awareness and increase sales.

e) Government Policies and Regulations

- Government policies influence business operations through taxation, subsidies, and regulations.
- Consumer protection laws ensure product quality and safety.
- Industrial policies support domestic production and economic development.

f) Infrastructure and Logistics

- Availability of roads, railways, ports, electricity, and storage facilities affects market efficiency.
- Proper logistics reduce transportation costs and ensure timely delivery.

g) Employment and Economic Growth

- Domestic markets create employment opportunities.
- Growth of industries contributes to national income and economic development.
- Strong domestic markets enhance industrial and agricultural productivity.

2. INTERNATIONAL MARKET (GLOBAL MARKET)

The international market involves trade and business activities between different countries. It enables the exchange of goods, services, technology, and capital across borders.

a) Export and Import Opportunities

- Countries export products in which they have a competitive advantage.
- Imports provide access to raw materials, machinery, and products unavailable domestically.
- International trade expands market opportunities and increases revenue.

b) Global Demand and Consumer Preferences

- Consumer needs differ across countries due to cultural, social, and economic factors.
- Businesses adapt products and marketing strategies according to foreign market requirements.
- Understanding international trends helps companies remain competitive.

c) Foreign Exchange and Currency Fluctuations

- International trade involves transactions in foreign currencies.
- Exchange rate fluctuations affect export earnings and import costs.
- Businesses manage currency risks through financial strategies.

d) Trade Policies and Agreements

- International trade is influenced by tariffs, quotas, customs duties, and trade regulations.
- Free trade agreements and regional economic organizations facilitate market access.
- Organizations such as World Trade Organization regulate global trade practices.

e) International Competition

- Companies face competition from multinational corporations and foreign producers.
- Competitive advantages include product quality, cost efficiency, innovation, and branding.
- Businesses invest in research and development to maintain competitiveness.

f) Transportation and Logistics

- International trade depends on shipping, air transport, ports, and supply chain management.
- Efficient logistics reduce costs and ensure timely delivery of products.
- Modern technologies improve tracking and inventory management.

g) Quality Standards and Certifications

- Products must comply with international quality and safety standards.
- Certifications such as ISO standards enhance product acceptance in foreign markets.
- Compliance with environmental and labor regulations is essential.

h) Market Risks and Challenges

- Political instability, economic recessions, and trade restrictions can affect international business.
- Cultural differences and language barriers require careful market planning.
- Global crises and supply chain disruptions may impact trade.

i) Economic Benefits

- International trade increases foreign exchange earnings.
- It promotes industrial development and technology transfer.
- Access to larger markets leads to economies of scale and increased profitability.
- It strengthens international relations and economic cooperation.

IMPORTANCE OF COMMERCIAL ASPECTS

1. Promote economic growth and industrial development.
2. Increase employment opportunities and income generation.
3. Improve market efficiency and resource utilization.
4. Enhance competitiveness and innovation.
5. Facilitate access to national and global markets.
6. Increase profitability and business sustainability.
7. Promote international trade and foreign exchange earnings.
8. Support balanced economic development and consumer welfare.



UNIT II

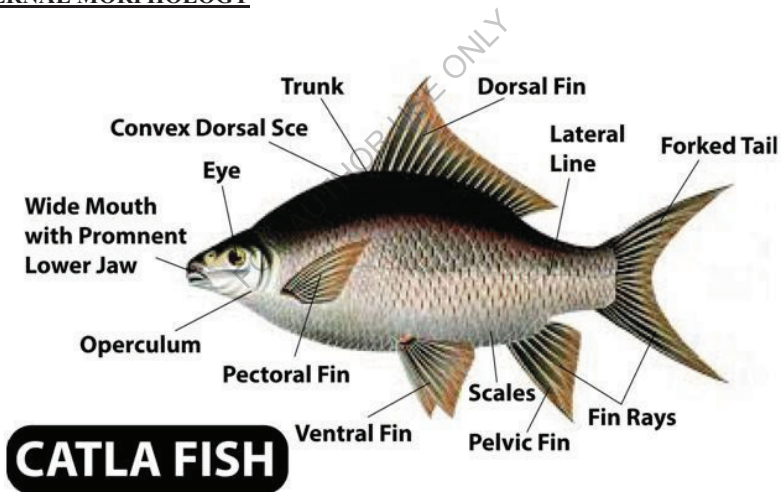
EXTERNAL MORPHOLOGY OF A FISH CATLA

Catla is one of the Indian major carps widely cultured in freshwater ponds, reservoirs, and rivers. It is a fast-growing fish and an important species in aquaculture.

Classification

- **Common Name:** Catla
- **Scientific Name:** *Catla catla*
- **Phylum:** Chordata
- **Class:** Actinopterygii
- **Order:** Cypriniformes
- **Family:** Cyprinidae

EXTERNAL MORPHOLOGY



1. Body Shape

- Body is deep, broad, and laterally compressed.
- The head is large and broad.
- Dorsal profile is more arched than the ventral profile.
- Body is streamlined, helping in efficient swimming.

2. Head

- Head is large and occupies a considerable portion of the body.
- Snout is short and blunt.
- Eyes are large and situated laterally on the head.
- Nostrils are present in front of the eyes.

3. Mouth

- Mouth is wide and terminal to slightly upward-directed.
- Lower jaw is prominent.
- Lips are thin.
- Barbels are absent.

4. Scales

- Body is covered with large cycloid scales.
- Scales are smooth, thin, and overlapping.
- They protect the body and reduce friction during swimming.

5. Fins

Catla possesses the following fins:

Paired Fins

- **Pectoral fins:** One pair located behind the gill covers; help in steering and balancing.
- **Pelvic fins:** One pair located on the ventral side; help in maintaining equilibrium.

Unpaired Fins

- **Dorsal fin:** Single dorsal fin on the back; helps maintain balance.
- **Anal fin:** Located behind the anus; assists in stabilization.
- **Caudal fin:** Large and deeply forked; provides propulsion during swimming.

6. Gill Cover (Operculum)

- Gill chambers are covered by bony opercula.
- Protects the delicate gills.
- Assists in respiration by regulating water flow.

7. Lateral Line

- A distinct lateral line runs along both sides of the body.
- Functions as a sensory organ to detect vibrations and movements in water.

8. Coloration

- Back is dark grey to blackish.
- Sides are silvery.
- Belly is whitish.
- Fins are generally grey or slightly dark.

Adaptive Features

- Streamlined body reduces water resistance.
- Forked caudal fin enables rapid swimming.
- Largemouth helps feed on plankton near the water surface.
- Lateral line helps detect predators and prey.

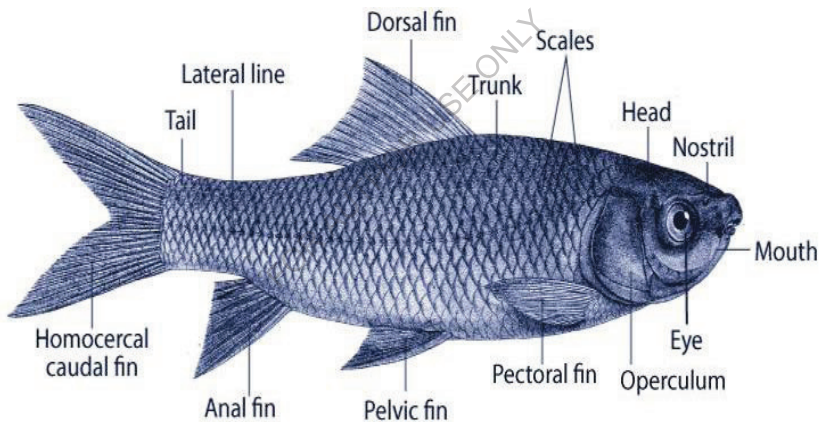


EXTERNAL MORPHOLOGY OF A FISH ROHU

A fish Rohu is one of the most important Indian major carps and is widely cultured in freshwater ponds, lakes, rivers, and reservoirs. It is valued for its rapid growth and high nutritional quality.

CLASSIFICATION

- **Common Name:** Rohu
- **Scientific Name:** *Labeo rohita*
- **Phylum:** Chordata
- **Class:** Actinopterygii
- **Order:** Cypriniformes
- **Family:** Cyprinidae



EXTERNAL MORPHOLOGY

1. Body Shape

- Body is elongated, moderately deep, and laterally compressed.
- Dorsal profile is more convex than the ventral profile.
- The body is streamlined, aiding efficient swimming.

2. Head

- Head is comparatively small to medium-sized.
- Snout is blunt and slightly projecting.

- Eyes are moderately large and located laterally.
- Nostrils are present in front of the eyes.

3. Mouth

- Mouth is small and subterminal (located slightly below the tip of the snout).
- Lips are thick and fringed.
- A pair of small rudimentary barbels may be present at the corners of the mouth.
- Adapted for feeding in the middle layers of water.

4. Scales

- Body is covered with large cycloid scales.
- Scales are smooth, thin, and overlapping.
- They protect the body and reduce friction while swimming.

5. Fins

Paired Fins

- **Pectoral fins:** One pair behind the operculum; help in steering and balance.
- **Pelvic fins:** One pair on the ventral side; maintain equilibrium.

Unpaired Fins

- **Dorsal fin:** Single dorsal fin located on the back; provides stability.
- **Anal fin:** Situated behind the anus; assists in balance.
- **Caudal fin:** Large and deeply forked; helps in fast swimming and propulsion.

6. Operculum (Gill Cover)

- Gills are protected by bony opercula.
- Helps regulate water flow over the gills during respiration.

7. Lateral Line

- A distinct lateral line extends from the operculum to the base of the caudal fin.
- Detects vibrations and water movements.

8. Coloration

- Back is dark bluish or brownish-grey.
- Sides are silvery.
- Belly is whitish.
- Fins are usually darker than the body.

ADAPTIVE FEATURES

- Streamlined body enables efficient swimming.
- Forked caudal fin provides speed and propulsion.
- Subterminal mouth is adapted for feeding in the middle water column.
- Lateral line helps detect prey and predators.

ECONOMIC IMPORTANCE

- One of the most important freshwater food fishes in India.
- Widely cultured in composite fish farming.
- Rich source of protein and nutrients.
- Contributes significantly to aquaculture production.

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EXTERNAL MORPHOLOGY OF A FISH MRIGAL

Mrigal is one of the three major Indian carps, along with Catla and Rohu. It is a freshwater fish widely cultured in ponds, reservoirs, and rivers throughout India.

CLASSIFICATION

- **Common Name:** Mrigal
- **Scientific Name:** *Cirrhinus mrigala*
- **Phylum:** Chordata
- **Class:** Actinopterygii
- **Order:** Cypriniformes
- **Family:** Cyprinidae

Mrigal



Identification characters:

- 1) Body elongated with depressed and obtusely rounded snout.
- 2) scales are large & Cycloid type
- 3) Mouth is terminal and wide, lips without fringe, upper lip not continuous with lower lip. One pair of short rostral barbules present in Mouth.
- 4) Caudal fin homo-cercal with deeply forked. Body colour is usually dark grey on the back and silvery on the sides and belly.

EXTERNAL MORPHOLOGY

1. Body Shape

- Body is elongated, streamlined, and laterally compressed.
- Dorsal and ventral profiles are moderately curved.
- The body shape is well adapted for swimming in freshwater habitats.

2. Head

- Head is comparatively small.
- Snout is blunt and rounded.
- Eyes are medium-sized and situated laterally.
- Nostrils are present in front of the eyes.

3. Mouth

- Mouth is subterminal (located below the tip of the snout).
- Lips are thick and fleshy.
- Barbels are absent.
- Mouth is adapted for bottom feeding.

4. Scales

- Body is covered with large cycloid scales.
- Scales are smooth, thin, and overlapping.
- They protect the body and reduce friction during movement.

5. Fins

Paired Fins

- **Pectoral fins:** One pair located behind the operculum; help in steering and balance.
- **Pelvic fins:** One pair on the ventral side; assist in maintaining equilibrium.

Unpaired Fins

- **Dorsal fin:** Single dorsal fin on the back; helps maintain stability.
- **Anal fin:** Located behind the anus; aids in balancing.
- **Caudal fin:** Large and deeply forked; provides propulsion for swimming.

6. Operculum (Gill Cover)

- Gills are covered by a bony operculum.
- Protects the gills and assists in respiration.

7. Lateral Line

- A distinct lateral line runs from the operculum to the base of the caudal fin.
- Functions as a sensory organ to detect vibrations and movements in water.

8. Coloration

- Back is dark grey or brownish.
- Sides are silvery.
- Belly is whitish.
- Fins are generally grey to dark in color.

ADAPTIVE FEATURES

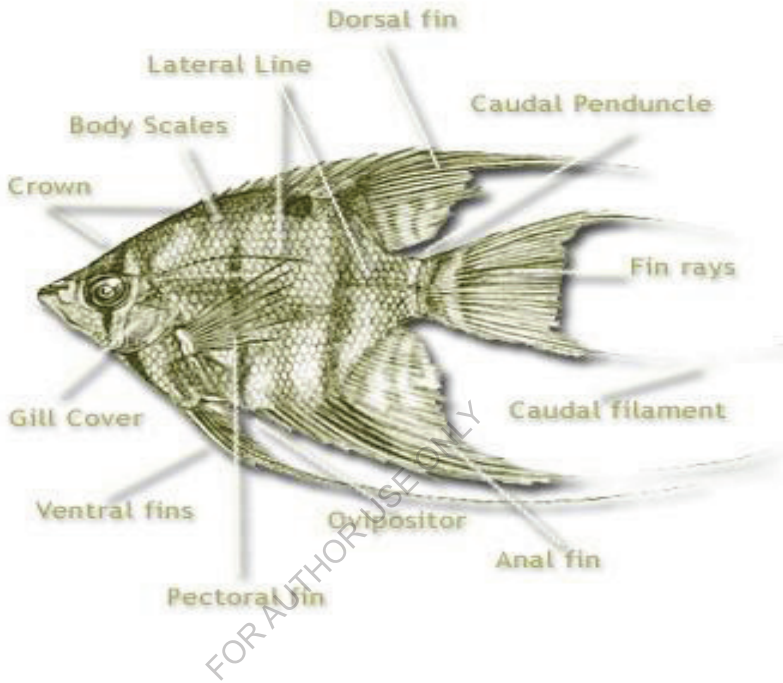
- Streamlined body reduces water resistance.
- Subterminal mouth is suitable for feeding on bottom-dwelling organisms and organic matter.
- Forked caudal fin helps in efficient swimming.
- Lateral line detects environmental changes and nearby organisms.

ECONOMIC IMPORTANCE

- Important freshwater food fish in India.
- Widely used in composite fish culture with Catla and Rohu.
- Fast-growing and highly valued in aquaculture.
- Rich source of protein and nutrients.

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EXTERNAL MORPHOLOGY OF A FISH ANGEL FISH



Freshwater Angelfish is a popular ornamental freshwater fish belonging to the family Cichlidae. It is native to the Amazon River basin of South America and is widely kept in aquariums due to its attractive appearance and graceful swimming behavior.

CLASSIFICATION

- **Common Name:** Angelfish
- **Scientific Name:** *Pterophyllum scalare*
- **Phylum:** Chordata
- **Class:** Actinopterygii
- **Order:** Cichliformes
- **Family:** Cichlidae

EXTERNAL MORPHOLOGY

1. Body Shape

- Body is highly compressed laterally and disc-shaped.
- Body is deep and thin, giving it a triangular appearance.
- This shape helps the fish move easily among aquatic plants.

2. Head

- Head is relatively small compared to body size.
- Forehead is slightly curved.
- Eyes are large and positioned on either side of the head.
- Nostrils are present in front of the eyes.

3. Mouth

- Mouth is small and terminal (located at the front of the head).
- Jaws contain small teeth used for feeding on small organisms.
- Adapted for capturing insects, larvae, and small aquatic animals.

4. Scales

- Body is covered with ctenoid scales.
- Scales are rough-edged and overlap one another.
- They provide protection and reduce water resistance.

5. Fins

Paired Fins

- **Pectoral fins:** Transparent and fan-shaped; used for steering and maneuvering.
- **Pelvic fins:** Long, thread-like fins hanging below the body; act as sensory organs and aid in balance.

Unpaired Fins

- **Dorsal fin:** Very long and pointed, extending from the middle of the back toward the tail.
- **Anal fin:** Long and pointed, mirroring the dorsal fin.
- **Caudal fin:** Fan-shaped tail fin used for propulsion.

6. Operculum (Gill Cover)

- Gills are protected by a bony operculum.
- Assists in respiration by regulating water flow over the gills.

7. Lateral Line

- A distinct lateral line is present on the sides of the body.
- Detects vibrations and movements in the surrounding water.

8. Coloration

- Body is usually silver with black vertical bands.
- Many ornamental varieties occur in gold, marble, black, koi, and albino colors.
- Color patterns provide camouflage and ornamental value.

ADAPTIVE FEATURES

- Deep, compressed body helps movement among aquatic vegetation.
- Long dorsal and anal fins improve stability and graceful swimming.
- Thread-like pelvic fins function as sensory structures.
- Lateral line helps detect prey, predators, and obstacles.

ECONOMIC IMPORTANCE

- One of the most popular ornamental aquarium fishes worldwide.
- High commercial value in the aquarium trade.
- Easy to breed and maintain under suitable conditions.

EXTERNAL MORPHOLOGY OF A FISH KOI FISH

Koi is an ornamental variety of the common carp. It is one of the most popular decorative fishes in the world and is widely cultured in ponds and water gardens for its bright colors and attractive patterns.

CLASSIFICATION

- Common Name: Koi Fish
- Scientific Name: *Cyprinus carpio*
- Phylum: Chordata
- Class: Actinopterygii
- Order: Cypriniformes
- Family: Cyprinidae

EXTERNAL MORPHOLOGY

1. Body Shape

- Body is elongated, spindle-shaped, and laterally compressed.
- The body is robust and streamlined.
- Shape is adapted for efficient swimming in ponds and lakes.

2. Head

- Head is broad and moderately large.
- Snout is blunt and rounded.
- Eyes are medium-sized and located laterally.
- Two pairs of nostrils are present in front of the eyes.

3. Mouth

- Mouth is terminal and protrusible (can extend outward during feeding).
- Two pairs of barbels are present around the mouth.
- Barbels act as sensory organs to locate food in muddy water.

4. Scales

- Body is covered with large cycloid scales.
- Scales are smooth, thin, and overlapping.
- Some ornamental varieties may have reduced scales or be scaleless.

5. Fins

Paired Fins

- Pectoral fins: One pair behind the gill cover; used for steering and balance.
- Pelvic fins: One pair on the ventral side; help maintain equilibrium.

Unpaired Fins

- Dorsal fin: Long dorsal fin extending along the back; provides stability.
- Anal fin: Located behind the anus; assists in balance.
- Caudal fin: Broad and forked tail fin; responsible for propulsion.

6. Operculum (Gill Cover)

- Gills are covered by a bony operculum.
- Protects the gills and assists in respiration.

7. Lateral Line

- A well-developed lateral line extends along both sides of the body.
- Detects vibrations, water currents, and nearby movements.

8. Coloration

Koi fish are famous for their bright and varied color patterns, including:

- White
- Red
- Orange
- Yellow
- Black
- Blue
- Cream

Different combinations of these colors form various koi varieties.

ADAPTIVE FEATURES

- Streamlined body enables smooth swimming.
- Barbels help locate food on the pond bottom.
- Protrusible mouth assists in sucking up food particles.
- Lateral line helps detect changes in the aquatic environment.

ECONOMIC IMPORTANCE

- Highly valued ornamental fish worldwide.
- Important in the aquarium and pond-fish industry.
- Symbolizes prosperity, luck, and perseverance in Japanese culture.
- Generates significant revenue through ornamental fish trade.



EXTERNAL MORPHOLOGY OF FIGHTER FISH

Siamese Fighting Fish, commonly known as the Fighter Fish or Betta Fish, is a popular ornamental freshwater fish. It is native to Southeast Asia and is renowned for its brilliant colors, long flowing fins, and territorial behavior.

CLASSIFICATION

- **Common Name:** Fighter Fish (Betta Fish)
- **Scientific Name:** *Betta splendens*
- **Phylum:** Chordata
- **Class:** Actinopterygii
- **Order:** Anabantiformes
- **Family:** Osphronemidae

EXTERNAL MORPHOLOGY

1. Body Shape

- Body is elongated, slender, and laterally compressed.
- The body is streamlined, allowing easy movement through water.
- Males generally have larger and more elaborate fins than females.

2. Head

- Head is small and pointed.
- Snout is short and slightly pointed.
- Eyes are large and positioned laterally.
- Nostrils are present in front of the eyes.

3. Mouth

- Mouth is terminal and slightly upturned.
- Jaws contain small teeth.
- Adapted for feeding on insects, larvae, and small aquatic organisms near the water surface.

4. Scales

- Body is covered with small ctenoid scales.
- Scales overlap and provide protection.
- The scales often exhibit a metallic sheen.

5. Fins

Paired Fins

- **Pectoral fins:** One pair located behind the operculum; used for maneuvering.
- **Pelvic fins:** Long and slender, situated on the ventral side; assist in balance and movement.

Unpaired Fins

- **Dorsal fin:** Located on the back and helps maintain stability.
- **Anal fin:** Long and well developed, extending along the ventral side.
- **Caudal fin:** Large, broad, and often brightly colored; used for propulsion and display.

6. Operculum (Gill Cover)

- Gills are protected by a bony operculum.
- During aggressive displays, the operculum is flared outward to appear larger and intimidate rivals.

7. Lateral Line

- A lateral line system is present but less distinct than in many larger fishes.
- Helps detect vibrations and water movements.

8. Coloration

- Exhibits a wide range of brilliant colors, including:
 - Red
 - Blue
 - Green
 - Purple
 - White
 - Yellow
 - Black
- Selective breeding has produced many ornamental color varieties.

9. Special Feature – Labyrinth Organ

- Fighter fish possess a **labyrinth organ**, located above the gills.
- This organ allows them to breathe atmospheric oxygen directly from the air.
- Enables survival in oxygen-poor waters.

ADAPTIVE FEATURES

- Bright coloration helps in courtship and territorial displays.
- Long fins aid in display and attraction of mates.
- Uprturned mouth is suitable for surface feeding.
- Labyrinth organ allows breathing of atmospheric air.
- Opercular flaring helps defend territory.

ECONOMIC IMPORTANCE

- One of the most popular ornamental aquarium fishes worldwide.
- Easy to maintain and breed in captivity.
- Important species in the ornamental fish trade.
- Appreciated for its beauty and unique behavior.



UNIT III

AQUARIUM PREPARATION AND MAINTENANCE- KINDS OF TANKS, TANK SETTING, BIOLOGICAL FILTER AND AERATION, WATER MANAGEMENT, PLANTING, LIGHTING AND FEEDS IN INSTITUTIONS.

An aquarium is an artificially maintained water tank used for keeping fishes, aquatic plants, and other aquatic organisms. Proper aquarium preparation and maintenance are essential for the health, growth, and survival of ornamental fishes. Aquariums are widely used in homes, educational institutions, research laboratories, and public displays.

1. KINDS OF TANKS

Aquariums can be classified based on size, material, and purpose.

A. BASED ON WATER TYPE

Freshwater Aquarium

- Contains freshwater fishes and plants.
- Easy to maintain.
- Most commonly used in homes and institutions.

Marine Aquarium

- Contains saltwater fishes and marine organisms.
- Requires specialized equipment and maintenance.

Brackish Water Aquarium

- Contains water with moderate salinity.
- Suitable for fishes adapted to estuarine conditions.

B. Based on Construction Material

Glass Tank

- Most common type.
- Provides clear visibility.
- Durable and easy to clean.

Acrylic Tank

- Lightweight and stronger than glass.
- Better insulation.
- More resistant to breakage.

C. Based on Purpose

Home Aquarium

- Decorative and recreational.

Display Aquarium

- Used in museums, aquaria, and public institutions.

Breeding Aquarium

- Used for fish breeding and rearing fry.

Quarantine Tank

- Used for isolating sick or newly introduced fishes.

2. TANK SETTING

Proper tank setup is essential before introducing fishes.

Step 1: Selection of Tank

- Choose suitable size depending on fish species and number.
- Place the tank on a strong and level surface.

Step 2: Cleaning

- Wash the tank thoroughly with clean water.
- Avoid detergents or chemicals.

Step 3: Substrate Arrangement

- Add gravel, sand, or pebbles at the bottom.
- Provides support for plants and beneficial bacteria.

Step 4: Decoration

- Place rocks, driftwood, caves, and ornamental structures.
- Provide shelter and improve appearance.

Step 5: Filling Water

- Use clean, dechlorinated water.
- Maintain appropriate temperature and pH.

Step 6: Installation of Equipment

- Filters
- Aerators
- Heaters (if required)
- Lighting system

Step 7: Acclimatization of Fish

- Introduce fish gradually.
- Avoid sudden temperature or water-quality changes.

3. BIOLOGICAL FILTER

A biological filter maintains water quality through beneficial microorganisms.

Functions

- Converts toxic ammonia into nitrites.
- Converts nitrites into less harmful nitrates.
- Maintains a healthy aquatic environment.

Components

- Filter media (gravel, sponge, ceramic rings).
- Beneficial nitrifying bacteria.

Types

Under-Gravel Filter

- Installed beneath the substrate.

Sponge Filter

- Suitable for breeding and nursery tanks.

Canister Filter

- Highly efficient for large aquariums.

Power Filter

- Commonly used in home aquariums.

Advantages

- Removes harmful waste.
- Improves water quality.
- Promotes fish health.

4. AERATION

Aeration supplies oxygen to aquarium water.

Importance

- Supports fish respiration.
- Enhances bacterial activity.
- Prevents oxygen depletion.

METHODS

Air Pump

- Pumps air into the water.

Air Stones

- Produce fine bubbles.

Water Pumps

- Increase water circulation.

Fountain Systems

- Improve oxygen exchange at the water surface.

Benefits

- Maintains dissolved oxygen levels.
- Reduces fish stress.
- Improves biological filtration efficiency.

5. WATER MANAGEMENT

Good water quality is essential for aquarium success.

Important Parameters

Temperature

- Generally maintained between 24°C and 28°C for tropical fishes.

pH

- Most freshwater fishes prefer pH 6.5–7.5.

Dissolved Oxygen

- Must be adequate for fish survival.

Ammonia and Nitrite

- Should remain near zero.

Nitrate

- Maintained at low levels through water changes.

Water Change

- Replace 20–30% of water every week.
- Removes accumulated waste and toxins.

Cleaning

- Remove uneaten food.
- Clean glass surfaces and filter media regularly.

6. PLANTING IN AQUARIUM

Aquatic plants improve aquarium aesthetics and water quality.

Importance

- Produce oxygen.
- Absorb carbon dioxide and nitrates.
- Provide shelter and breeding sites.
- Enhance beauty.

Common Aquarium Plants

- Hydrilla
- Vallisneria
- Cabomba
- Amazon Sword
- Java Fern

Planting Procedure

- Place substrate at the bottom.
- Insert plant roots carefully.
- Provide adequate light and nutrients.

7. LIGHTING

Lighting is essential for plant growth and fish display.

Functions

- Supports photosynthesis.
- Enhances fish coloration.
- Maintains natural day-night cycles.

TYPES OF LIGHTING

LED Lights

- Energy-efficient and long-lasting.

Fluorescent Lights

- Commonly used in aquariums.

Full-Spectrum Lights

- Ideal for planted tanks.

Lighting Duration

- 8–12 hours per day.
- Excessive lighting may cause algal growth.

8. FEEDS IN INSTITUTIONS

Institutions such as colleges, research centers, and ornamental fish units provide balanced nutrition.

Natural Feeds

- Plankton
- Algae
- Aquatic insects
- Worms

Live Feeds

- Tubifex worms
- Daphnia
- Artemia (Brine shrimp)

Artificial Feeds

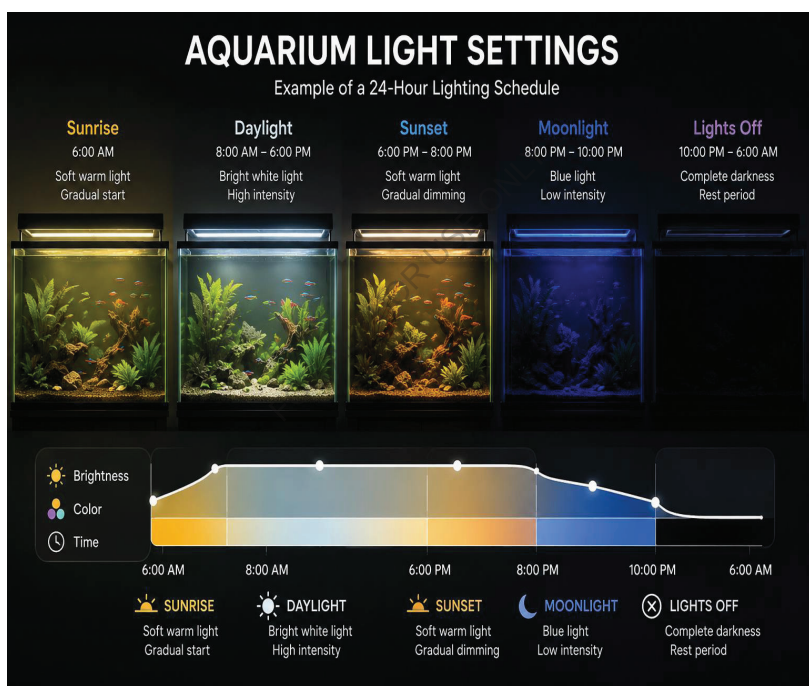
- Pellets
- Flakes
- Granules

Feeding Schedule

- Feed 2–3 times daily.
- Provide only the amount consumed within a few minutes.
- Avoid overfeeding to prevent water pollution.

Maintenance Practices

1. Daily observation of fish health.
2. Regular feeding and cleaning.
3. Weekly partial water changes.
4. Checking filter and aeration systems.
5. Monitoring temperature and pH.
6. Removal of dead plants and wastes.
7. Disease prevention and quarantine of new fishes.



UNIT IV

LIVE FISH TRANSPORT – HANDLING AND FEEDING

Live fish transport is the process of moving fish from one place to another while keeping them alive, healthy, and free from stress. Proper handling and feeding practices are essential to minimize mortality and ensure successful transportation in aquaculture and ornamental fish culture.

1. HANDLING OF LIVE FISH

Fish are delicate animals and require careful handling to avoid injury and stress.

Importance of Proper Handling

- Prevents physical injuries.
- Reduces stress and mortality.
- Maintains fish health and quality.
- Improves survival during transport.

Handling Procedures

A. Before Handling

- Select healthy and disease-free fish.
- Avoid handling during extreme temperatures.
- Use clean equipment and containers.

B. During Handling

- Handle fish gently.
- Avoid overcrowding.
- Use soft nets to reduce scale and fin damage.
- Keep fish moist at all times.
- Minimize exposure to air.

C. After Handling

- Transfer fish carefully into holding tanks.
- Observe fish for signs of stress or injury.
- Provide good water quality and aeration.

2. LIVE FISH TRANSPORT

Transport may be short-distance or long-distance depending on the purpose.

OBJECTIVES

- Deliver fish safely.
- Minimize stress and mortality.
- Maintain fish quality and health.

TYPES OF TRANSPORT

A. Open Transport System

- Fish are transported in tanks, tubs, or containers.
- Suitable for short distances.
- Requires continuous aeration.

B. Closed Transport System

- Fish are packed in sealed plastic bags containing water and oxygen.
- Suitable for long-distance transport.
- Commonly used for ornamental fishes and fish seed.

TRANSPORT CONTAINERS

Plastic Bags

- Widely used for ornamental fishes.
- Filled with one-third water and two-thirds oxygen.

Fiberglass Tanks

- Used for transporting large numbers of fish.

Plastic or Metal Tanks

- Suitable for short-distance transport.

Oxygenated Containers

- Equipped with oxygen supply systems.

FACTORS AFFECTING FISH TRANSPORT

Temperature

- Maintain suitable temperature according to fish species.
- High temperatures increase stress and oxygen demand.

Dissolved Oxygen

- Adequate oxygen is essential for survival.
- Oxygen cylinders or aerators may be used.

Water Quality

- Clean, disease-free water should be used.
- Maintain suitable pH and low ammonia levels.

Stocking Density

- Avoid overcrowding.
- Higher density increases stress and mortality.

Duration of Transport

- Longer transport requires better oxygenation and monitoring.

3. FEEDING BEFORE AND AFTER TRANSPORT

Proper feeding management is important for successful fish transportation.

Feeding Before Transport

Fasting

- Fish are usually starved before transport.
- Fasting period:
 - Fry: 12–24 hours
 - Fingerlings: 24 hours
 - Adult fish: 24–48 hours

Advantages of Fasting

- Reduces waste production.
- Lowers ammonia accumulation.
- Improves water quality.
- Decreases oxygen consumption.
- Reduces transport stress.

Feeding After Transport

Recovery Period

- Allow fish to recover after transport.
- Do not feed immediately.

Feeding Schedule

- Feed fish after 12–24 hours of acclimatization.
- Start with small quantities.
- Gradually return to normal feeding rates.

Types of Feed

- Live feed (Artemia, Daphnia, Tubifex).
- Commercial pellets.
- Flake feeds for ornamental fishes.

STRESS MANAGEMENT DURING TRANSPORT

- Maintain adequate oxygen levels.
- Avoid sudden temperature changes.
- Reduce handling frequency.
- Use clean transport containers.
- Keep transport duration as short as possible.
- Acclimatize fish slowly before release into new water.

PRECAUTIONS

1. Use healthy fish only.
2. Fast fish before transport.
3. Avoid overcrowding.
4. Provide sufficient oxygen.
5. Maintain proper temperature.
6. Handle fish gently.
7. Monitor water quality regularly.
8. Acclimatize fish before stocking.

FISH DISEASES AND THEIR CONTROL – VIRAL HEMORRHAGIC SEPTICEMIA (VHS)

Viral Hemorrhagic Septicemia (VHS) is a serious and highly contagious viral disease affecting many freshwater and marine fish species. It causes internal bleeding (hemorrhage), anemia, and high mortality, particularly in young fish. The disease is of major economic importance in aquaculture and fisheries worldwide.

CAUSATIVE AGENT

- Caused by **Viral Hemorrhagic Septicemia Virus (VHSV)**.
- The virus belongs to the family Rhabdoviridae and genus *Novirhabdovirus*.
- It can infect a wide range of fish species.

Susceptible Fish Species

- Trout
- Salmon
- Pike
- Herring
- Cod
- Turbot
- Many other freshwater and marine fishes

Transmission

The virus spreads through:

1. Direct contact with infected fish.
2. Contaminated water.
3. Infected eggs and fish products.
4. Equipment, nets, and transport containers.
5. Carrier fish that show no visible symptoms.

Clinical Signs (Symptoms)

External Symptoms

- Darkening of body coloration.
- Hemorrhages (red spots) on skin and fins.
- Bulging eyes (exophthalmia).
- Pale gills due to anemia.

- Distended abdomen.
- Erratic swimming behavior.
- Lethargy and loss of appetite.

Internal Symptoms

- Hemorrhages in muscles and internal organs.
- Enlarged spleen and kidney.
- Fluid accumulation in body cavity.
- Liver degeneration.

DIAGNOSIS

- Observation of clinical signs.
- Laboratory examination.
- Virus isolation in cell culture.
- Molecular techniques such as PCR.
- Serological tests.

ECONOMIC IMPORTANCE

- Causes heavy mortality in fish farms.
- Reduces fish production and profitability.
- Leads to trade restrictions and quarantine measures.
- Results in significant economic losses to the aquaculture industry.

CONTROL AND PREVENTION

1. Biosecurity Measures

- Use disease-free fish and eggs.
- Restrict movement of infected fish.
- Maintain strict farm hygiene.

2. Quarantine

- Isolate newly introduced fish.
- Prevent contact between healthy and infected stocks.

3. Disinfection

- Regularly disinfect tanks, nets, equipment, and transport containers.
- Remove dead fish immediately.

4. Water Quality Management

- Maintain good water quality.
- Avoid overcrowding.
- Reduce environmental stress on fish.

5. Surveillance and Monitoring

- Conduct regular health checks.
- Report disease outbreaks promptly.

6. Vaccination

- Vaccines are available in some regions and for certain fish species.
- Vaccination helps reduce disease occurrence and mortality.

TREATMENT

- There is **no specific cure** for Viral Hemorrhagic Septicemia.
- Antibiotics are ineffective because VHS is caused by a virus.
- Management focuses on prevention, biosecurity, and control of disease spread.

PREVENTION SUMMARY

- Maintain disease-free broodstock.
- Practice strict sanitation.
- Quarantine new fish.
- Monitor fish health regularly.
- Avoid stress and overcrowding.
- Dispose of infected fish properly.

FISH DISEASES AND THEIR CONTROL – BACTERIAL (FIN ROT)

Fin Rot is one of the most common bacterial diseases affecting both ornamental and cultured fishes. It primarily attacks the fins and tail, causing gradual deterioration of fin tissues. The disease is usually associated with poor water quality, overcrowding, stress, and physical injury.



CAUSATIVE AGENT

Fin Rot is mainly caused by bacteria such as:

- *Aeromonas hydrophila*
- *Pseudomonas fluorescens*
- *Flavobacterium columnare*

These bacteria are often present in water and infect fish when their resistance is lowered.

SUSCEPTIBLE FISH SPECIES

- Goldfish
- Koi
- Betta (Fighter fish)
- Angelfish
- Guppies
- Carp and other freshwater fishes

CAUSES OF FIN ROT

1. Poor water quality.
2. Overcrowding in tanks or ponds.
3. Physical injury to fins.

4. Stress due to sudden environmental changes.
5. Inadequate nutrition.
6. Presence of pathogenic bacteria.

SYMPTOMS

External Symptoms

- Fraying and ragged fin edges.
- White, red, or black discoloration on fins.
- Gradual shortening of fins and tail.
- Inflamed fin bases.
- Loss of fin tissue.
- Reduced swimming activity.
- Loss of appetite in severe infections.

Advanced Stage

- Fin tissue may completely erode.
- Secondary infections may develop.
- Fish become weak and susceptible to other diseases.

Diagnosis

- Visual observation of damaged fins.
- Examination of water quality.
- Laboratory identification of bacteria when necessary.

CONTROL AND PREVENTION

1. Maintain Good Water Quality

- Regular water changes.
- Proper filtration and aeration.
- Maintain suitable pH and temperature.
- Remove uneaten food and waste.

2. Reduce Stress

- Avoid overcrowding.
- Handle fish carefully.
- Maintain stable environmental conditions.

3. Proper Nutrition

- Provide balanced and nutritious feed.
- Avoid feeding spoiled food.

4. Quarantine

- Isolate infected fish immediately.
- Quarantine newly purchased fish before introducing them into the aquarium.

5. Sanitation

- Clean tanks, nets, and equipment regularly.
- Remove dead or diseased fish promptly.

TREATMENT

Salt Treatment

- Common aquarium salt may help in mild infections.
- Used under recommended dosages.

Antibacterial Treatment

- Suitable antibacterial medications may be used under professional guidance.
- Follow manufacturer instructions carefully.

Water Quality Improvement

- Often the most important step in recovery.
- Clean, well-oxygenated water helps damaged fins regenerate.

PREVENTIVE MEASURES

1. Maintain excellent water quality.
2. Avoid overcrowding.
3. Feed balanced diets.
4. Quarantine new fish.
5. Prevent injuries during handling.
6. Inspect fish regularly for early signs of disease.

ECONOMIC IMPORTANCE

- Causes reduced growth and poor appearance of ornamental fishes.
- Decreases market value.
- May lead to mortality if untreated.
- Results in economic losses in ornamental fish culture and aquaculture.

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FISH DISEASES AND THEIR CONTRO- FUNGAL (SAPROLEGNIASIS) DISEASES



Saprolegniasis is a common fungal disease of freshwater fishes caused by water molds belonging to the genus *Saprolegnia*. It affects fish of all ages and is characterized by cotton-like fungal growth on the skin, fins, gills, and eggs. The disease is especially common in fish that are injured, stressed, or maintained in poor water conditions.

CAUSATIVE AGENT

- Caused mainly by species of *Saprolegnia*.
- These fungi are naturally present in freshwater environments.
- They become pathogenic when fish are weakened or injured.

SUSCEPTIBLE FISH SPECIES

- Carp (Catla, Rohu, Mrigal)
- Goldfish
- Koi fish
- Angelfish
- Betta (Fighter fish)
- Trout and Salmon
- Fish eggs and fry

CAUSES AND PREDISPOSING FACTORS

1. Physical injuries to skin and fins.
2. Poor water quality.
3. Low water temperature.
4. Overcrowding.
5. Stress during handling and transport.
6. Secondary infection following bacterial or parasitic diseases.
7. Dead organic matter in ponds or aquariums.

SYMPTOMS

External Symptoms

- White or grey cotton-like patches on the body.
- Fungal growth on fins, skin, gills, and mouth.
- Damaged scales and skin lesions.
- Frayed fins.
- Excess mucus secretion.
- Loss of appetite.
- Lethargic behavior and reduced swimming activity.

In Fish Eggs

- White cotton-like fungal growth on eggs.
- Egg mortality and poor hatching rates.

PATHOGENESIS

- The fungus first attacks damaged or weakened tissues.
- Fungal threads (hyphae) penetrate the skin and underlying tissues.
- Severe infections can interfere with respiration and osmoregulation, leading to death.

DIAGNOSIS

- Observation of characteristic cotton-like growth.
- Microscopic examination of fungal hyphae.
- Laboratory identification when required.

CONTROL AND PREVENTION

1. Maintain Good Water Quality

- Regular water changes.
- Adequate filtration and aeration.
- Proper pH and temperature management.

2. Prevent Injury

- Handle fish carefully.
- Avoid rough nets and overcrowding.

3. Proper Sanitation

- Remove dead fish and organic debris.
- Clean tanks, ponds, and equipment regularly.

4. Quarantine

- Isolate infected fish.
- Observe newly introduced fish before stocking.

5. Reduce Stress

- Maintain suitable environmental conditions.
- Provide balanced nutrition.

TREATMENT

Salt Treatment

- Salt baths may help control mild fungal infections.

Antifungal Chemicals

- Approved antifungal treatments may be used under professional guidance.
- Treatment should follow recommended dosages and safety guidelines.

Egg Treatment

- In hatcheries, infected eggs are removed promptly.
- Good water circulation and hygiene help prevent fungal spread.

PREVENTIVE MEASURES

1. Maintain clean water and proper aeration.
2. Avoid overcrowding.
3. Prevent injuries during handling.

4. Provide nutritious feed.
5. Remove infected fish and eggs immediately.
6. Disinfect equipment regularly.

ECONOMIC IMPORTANCE

- Causes mortality in fish and fish eggs.
- Reduces growth and market value of ornamental fishes.
- Leads to losses in hatcheries and aquaculture farms.
- Increases management and treatment costs.



UNIT V
BREEDING – COMMON CHARACTERS AND SEXUAL DIMORPHISM OF
FRESHWATER FISHES

Breeding is the process by which fishes reproduce and produce offspring. Successful breeding in freshwater fishes depends on the identification of males and females, understanding their reproductive behaviour, and recognizing the external differences between the sexes, known as **sexual dimorphism**.

COMMON CHARACTERS OF BREEDING FRESHWATER FISHES

During the breeding season, freshwater fishes exhibit several common reproductive characteristics.

1. Maturity of Gonads

- Testes in males and ovaries in females become fully developed.
- Breeding fish are known as brooders or broodstock.

2. Spawning Behavior

- Fishes become more active during the breeding period.
- Males often chase females before spawning.
- Courtship behavior is commonly observed.

3. Color Changes

- Breeding fishes may develop brighter coloration.
- Colors become more intense, especially in males.

4. Nest Building

- Some species construct nests for egg laying.
- Example: Fighter fish (*Betta splendens*) builds bubble nests.

5. Egg Production

- Females produce large numbers of eggs.
- Egg size and number vary among species.

6. Fertilization

- Most freshwater fishes show external fertilization.
- Eggs and sperm are released into water during spawning.

7. Parental Care

- Some species protect eggs and young ones.
- Others leave eggs unattended after spawning.

SEXUAL DIMORPHISM IN FRESHWATER FISHES

Sexual dimorphism refers to the visible differences between male and female fishes of the same species.

Importance

- Helps identify males and females for breeding.
- Essential in broodstock management.
- Improves breeding success in aquaculture and ornamental fish culture.

GENERAL DIFFERENCES BETWEEN MALE AND FEMALE FISHES

Character	Male Fish	Female Fish
Body Size	Usually smaller and slender	Usually larger and broader
Abdomen	Flat or narrow	Rounded and swollen during breeding
Coloration	Brighter and more attractive	Comparatively dull
Activity	More active and aggressive	Less active
Genital Opening	Small and narrow	Larger and prominent
Breeding Tubercles	Present in some species	Usually absent

SEXUAL DIMORPHISM IN COMMON FRESHWATER FISHES

1. Rohu (*Labeo rohita*)

Rohu

Male

- Slender body.
- Rough pectoral fins during breeding.
- Releases milt when gentle pressure is applied.

Female

- Larger and heavier body.
- Soft, swollen abdomen filled with eggs.
- Releases eggs when mature.

2. Catla (*Catla catla*)

Catla

Male

- Smaller and slender.
- Rough pectoral fin surface.
- Milt oozes on gentle pressure.

Female

- Larger body size.
- Bulging abdomen during breeding season.
- Genital opening becomes reddish and enlarged.

3. Mrigal (*Cirrhinus mrigala*)

Mrigal

Male

- Slender body.
- Rough pectoral fins.
- Produces milt during breeding season.

Female

- Rounded abdomen.
- Larger and heavier than males.
- Contains mature eggs.

4. Koi Fish (*Cyprinus carpio*)

Koi

Male

- More slender body.
- Breeding tubercles appear on gill covers and pectoral fins.
- More active during spawning.

Female

- Fuller and broader abdomen.
- Larger body during egg development.

5. Fighter Fish (*Betta splendens*)

Siamese Fighting Fish

Male

- Bright coloration.
- Long flowing fins.
- Highly territorial.
- Builds bubble nests.

Female

- Shorter fins.
- Less colorful.
- More rounded abdomen.

6. Angelfish (*Pterophyllum scalare*)

Freshwater Angelfish

Male

- Slightly larger forehead.
- More pointed breeding tube.

Female

- Broader abdomen during breeding.
- Blunt and wider ovipositor for egg laying.

SIGNIFICANCE OF SEXUAL DIMORPHISM IN FISH BREEDING

1. Helps in selecting broodstock.
2. Facilitates artificial breeding programs.
3. Improves hatchery management.
4. Assists in maintaining breeding ratios.
5. Enhances seed production in aquaculture.

LIVE FISH TRANSPORT

Live fish transport is the process of moving live fishes from one location to another while maintaining their health, survival, and quality. It is an important activity in aquaculture, ornamental fish culture, hatcheries, and fish marketing. Proper transport minimizes stress, injury, and mortality.

OBJECTIVES OF LIVE FISH TRANSPORT

1. To transport fish safely with minimum mortality.
2. To maintain fish health and quality.
3. To reduce stress during transportation.
4. To supply fish seed, broodstock, and ornamental fishes to different locations.
5. To support aquaculture and fish trade.

PRINCIPLES OF LIVE FISH TRANSPORT

- Provide adequate oxygen.
- Maintain good water quality.
- Control temperature.
- Avoid overcrowding.
- Minimize handling stress.
- Use suitable transport containers.

TYPES OF LIVE FISH TRANSPORT

1. Open Transport System

- Fish are transported in open containers such as tanks, tubs, and barrels.
- Continuous aeration is provided.
- Suitable for short-distance transport.

Advantages

- Easy monitoring of fish.
- Simple handling.

Disadvantages

- Higher risk of oxygen depletion.
- Water quality deteriorates quickly.

2. Closed Transport System

- Fish are packed in sealed plastic bags containing water and oxygen.
- Commonly used for ornamental fishes and fish seed.
- Suitable for long-distance transportation.

Advantages

- Better oxygen availability.
- Lower risk of contamination.
- Convenient for long journeys.

Disadvantages

- Limited space.
- Requires proper packing techniques.

TRANSPORT CONTAINERS**Plastic Bags**

- Most common method for ornamental fish transport.
- Bag contains:
 - One-third water
 - Two-thirds oxygen

Plastic Containers

- Used for short-distance transportation.

Fiberglass Tanks

- Suitable for transporting large numbers of fish.

Metal Tanks

- Used in specialized transport vehicles.

PREPARATION OF FISH BEFORE TRANSPORT**Selection of Fish**

- Choose healthy and disease-free fish.
- Avoid injured or weak fish.

Fasting

- Fish are starved before transport.
- Fry: 12–24 hours.
- Fingerlings: 24 hours.
- Adult fish: 24–48 hours.

Benefits of Fasting

- Reduces waste production.
- Lowers ammonia levels.
- Improves water quality.
- Reduces oxygen demand.

PACKING OF FISH**Steps**

1. Fill the transport bag with clean water.
2. Introduce the required number of fish.
3. Fill the remaining space with oxygen.
4. Seal the bag securely.
5. Place bags in insulated boxes for protection.

FACTORS AFFECTING FISH TRANSPORT

Temperature

- Maintain optimum temperature.
- High temperatures increase oxygen consumption and stress.

Dissolved Oxygen

- Adequate oxygen is essential for fish survival.
- Oxygen cylinders or aerators may be used.

Water Quality

- Water should be clean and free from pollutants.
- Maintain suitable pH and low ammonia levels.

Stocking Density

- Avoid overcrowding.
- Density depends on species, size, and transport duration.

Transport Duration

- Longer transport requires greater attention to oxygen and water quality.

Care During Transport

- Monitor fish behavior regularly.
- Avoid sudden shaking and vibrations.
- Protect containers from direct sunlight.
- Maintain proper aeration.
- Replace water if necessary, during long journeys.

Acclimatization After Transport Before releasing fish into ponds or aquariums:

1. Float the transport bag in the new water for 15–30 minutes.
2. Allow temperatures to equalize.
3. Gradually mix small amounts of pond or aquarium water into the bag.
4. Release fish gently.

Importance

- Prevents temperature shock.
- Reduces stress.
- Improves survival rate.

Precautions

1. Use healthy fish only.
2. Fast fish before transport.
3. Avoid overcrowding.
4. Provide adequate oxygen.
5. Maintain good water quality.
6. Handle fish gently.
7. Protect fish from extreme temperatures.
8. Acclimatize fish properly after transport.

ADVANTAGES OF PROPER LIVE FISH TRANSPORT

- High survival rate.
- Reduced stress and disease.
- Better growth after stocking.
- Improved fish quality.
- Increased profitability in aquaculture and ornamental fish trade.

BUDGET FOR SETTING UP AN AQUARIUM FISH UNIT

The cost of setting up an aquarium depends on the size of the tank, equipment, fish species, and decorations. The following is a sample budget for a small freshwater ornamental aquarium (about 2–3 feet tank) suitable for educational institutions or home use.

Item	Approximate Cost (₹)
Glass Aquarium Tank (60–90 L)	2,000 – 4,000
Aquarium Stand	1,500 – 3,000
Filter (Internal/Power Filter)	800 – 2,000
Air Pump and Air Stone	500 – 1,200
LED Lighting	500 – 1,500
Gravel/Substrate	500 – 1,500
Aquatic Plants	300 – 1,000
Decorative Materials (rocks, ornaments)	500 – 2,000
Water Conditioner	200 – 500
Fish Feed	200 – 500
Ornamental Fish (10–15 fishes)	500 – 3,000
Nets, Thermometer, Miscellaneous Items	300 – 1,000

Total Estimated Cost

₹7,800 – ₹20,200

MONTHLY MAINTENANCE COST

Item	Approximate Cost (₹/month)
Fish Feed	100 – 300
Electricity (filter, air pump, lights)	150 – 500
Water Conditioner/Medicines	50 – 200
Plant and Equipment Maintenance	100 – 300

Total Monthly Cost

₹400 – ₹1,300 per month

BUDGET FOR AN INSTITUTIONAL AQUARIUM UNIT

For schools, colleges, or laboratories maintaining multiple tanks:

Item	Approximate Cost (₹)
5–10 Aquariums	20,000 – 50,000
Filtration and Aeration Systems	5,000 – 15,000
Aquarium Stands	5,000 – 20,000
Ornamental Fish Stock	5,000 – 20,000
Plants and Decorations	3,000 – 10,000
Miscellaneous Equipment	2,000 – 10,000

Total Institutional Setup Cost

₹40,000 – ₹1,25,000 (depending on scale and quality)

FACTORS AFFECTING THE BUDGET

1. Aquarium size and material (glass or acrylic).
2. Type of filtration and aeration system.
3. Number and species of fish.
4. Use of live plants and decorations.
5. Lighting quality.
6. Maintenance and electricity costs.

SELF-EMPLOYMENT OPPORTUNITIES IN AQUARIUM FISH KEEPING

The ornamental fish industry is one of the fastest-growing sectors of aquaculture. Aquarium fish keeping can provide excellent self-employment opportunities for individuals, youth, women entrepreneurs, and small-scale farmers. Public aquariums also create employment and educational opportunities while promoting awareness about aquatic life.

SELF-EMPLOYMENT OPPORTUNITIES IN AQUARIUM FISH KEEPING

1. Ornamental Fish Breeding

- Breeding ornamental fishes such as:
 - Koi
 - Siamese Fighting Fish
 - Freshwater Angelfish
 - Goldfish
 - Guppies
- Sale of fry, fingerlings, and broodstock.
- Requires small investment and limited space.

2. Aquarium Fish Rearing

- Raising ornamental fishes for local markets.
- Supplying fish to aquarium shops and hobbyists.
- Can be operated from home.

3. Aquarium Shop Business

- Sale of ornamental fishes.
- Sale of aquariums, filters, aerators, plants, and accessories.
- Provides continuous income throughout the year.

4. Aquarium Fabrication

- Manufacturing glass aquariums of different sizes.
- Designing custom-made tanks for homes, offices, schools, and hotels.

5. Aquarium Maintenance Service

- Cleaning and maintaining aquariums in homes and institutions.
- Monthly maintenance contracts provide regular income.

6. Aquatic Plant Cultivation

- Production and sale of aquatic plants.
- Supply to aquarium shops and hobbyists.

7. Fish Feed Production

- Preparation and sale of ornamental fish feeds.
- Includes flake feeds, pellets, and live feeds.

8. Ornamental Fish Export

- Export of ornamental fishes to national and international markets.
- Generates higher profits and foreign exchange earnings.

Home-Based Aquarium Enterprise

Requirements

- Small room or covered area.
- Aquarium tanks.
- Aeration and filtration systems.
- Reliable water supply.
- Electricity.
- Fish feed and breeding materials.

Advantages

- Low initial investment.
- Can be managed as a part-time business.
- Suitable for rural and urban areas.
- Good market demand.

NEEDS AND IMPORTANCE OF PUBLIC AQUARIUMS

A public aquarium is a large facility where aquatic organisms are displayed for education, research, recreation, and conservation.

Educational Needs

- Provides knowledge about aquatic biodiversity.
- Helps students learn fish biology and ecology.
- Creates awareness about conservation.

Research Needs

- Supports scientific studies on aquatic organisms.
- Provides facilities for breeding and behavioral research.

Conservation Needs

- Conserves endangered fish species.
- Promotes awareness of aquatic ecosystem protection.

Recreational Needs

- Provides entertainment and relaxation.
- Attracts tourists and visitors.

Economic Needs

- Generates employment opportunities.
- Supports tourism and local businesses.
- Increases revenue through entry fees and related services.

FACILITIES REQUIRED FOR PUBLIC AQUARIUMS

1. Large display tanks.
2. Filtration and water-treatment systems.
3. Aeration systems.
4. Lighting facilities.
5. Quarantine and breeding units.
6. Fish feed storage facilities.
7. Educational display boards.
8. Trained staff and technicians.

EMPLOYMENT OPPORTUNITIES IN PUBLIC AQUARIUMS

- Aquarium Manager
- Aquarist
- Fish Breeder
- Aquarium Technician
- Fish Health Specialist
- Tour Guide
- Maintenance Staff
- Sales and Marketing Personnel

ADVANTAGES OF AQUARIUM-BASED SELF-EMPLOYMENT

1. Low investment compared to many businesses.
2. High demand for ornamental fishes.
3. Scope for expansion.
4. Generates regular income.
5. Suitable for small-scale entrepreneurs.
6. Promotes entrepreneurship and employment.



REFERENCES

1. The Simple Guide to Freshwater Aquariums — by David E. Boruchowitz ,Beginner-friendly introduction to setting up and maintaining freshwater aquariums.
2. The Complete Aquarium Guide — by Peter W. Scott ,Covers aquarium design, fish selection, water quality, and maintenance.
3. Encyclopedia of Aquarium & Pond Fish,Extensive fish species profiles with photographs and care requirements.
4. Aquarium Fish — by Dick Mills ,Identification and care of hundreds of freshwater fish species.
5. The 101 Best Freshwater Nano Species — by Mark Denaro ,Excellent for small aquarium enthusiasts.
6. The Conscientious Marine Aquarist — by Robert M. Fenner .One of the most recommended books for marine aquarium beginners and intermediate hobbyists.
7. The Reef Aquarium — by J. Charles Delbeek and Julian Sprung ,Considered a classic reference for reef aquarium management.
8. Ecology of the Planted Aquarium — by Diana Walstad,Scientifically grounded approach to planted aquariums.
9. Nature Aquarium World — by Takashi Amano ,Influential work on natural aquascaping techniques.
10. Manual of Fish Health — by Chris Andrews, Adrian Exell, and Neville Carrington Useful for diagnosing and treating fish diseases.
11. Aquarium Atlas — by Hans A. Baensch and Riehl Rüdiger ,Comprehensive species identification and husbandry information.
12. Boruchowitz, D. E. (2009). *The simple guide to freshwater aquariums*. T.F.H. Publications
13. Fenner, R. M. (2001). *The conscientious marine aquarist*. T.F.H. Publications.
14. Walstad, D. (2023). *Ecology of the planted aquarium* (3rd ed.). Echinodorus Publishing.
15. The Complete Aquarium — Peter W. Scott
16. The New Encyclopedia of the Aquarium — Peter W. Scott
17. Aquarium Atlas Volume 1 — Hans A. Baensch & Rüdiger Riehl. This volume covers aquarium setup, fish identification, aquatic plants, and fish care.

18. Aquarium Atlas Volume 2 — Hans A. Baensch & Rüdiger Riehl. Contains information on hundreds of additional fish and plant species.
19. Encyclopedia of Aquarium Plants — Peter Hiscock
20. Aquarium Plants Manual — Ines Scheurmann
21. The New Marine Aquarium — Michael S. Paletta
22. Natural Reef Aquariums — John H. Tullock
23. Marine Reef Aquarium Handbook — Robert J. Goldstein
24. The Saltwater Aquarium Handbook — George Blasiola
25. Ecology of the Planted Aquarium — Diana Walstad. Considered one of the most influential books on planted aquariums and aquatic ecology.
26. Nature Aquarium World — Takashi Amano
27. Aquascaping — George Farmer. Frequently recommended alongside Walstad and plant references by aquarium hobbyists.
28. Manual of Fish Health — Chris Andrews, Adrian Exell & Neville Carrington
29. Fish Disease: Diagnosis and Treatment — Edward J. Noga
30. Freshwater Aquariums For Dummies — Maddy Hargrove & Mic Hargrove
31. Aquarium Care of Cichlids — Mary Bailey & Peter Burgess
32. The Encyclopedia of Tropical Fishes — Dick Mills
33. Baensch, H. A., & Riehl, R. (1997). *Aquarium Atlas* (Vol. 1). Microcosm Ltd.
34. Walstad, D. L. (1999). *Ecology of the Planted Aquarium: A Practical Manual and Scientific Treatise for the Home Aquarist*. Echinodorus Publishing.
35. Fenner, R. M. (1996). *The Conscientious Marine Aquarist*. T.F.H. Publications.

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